Inf Technol Manag (2013) 14:143–157
DOI 10.1007/s10799-012-0126-9

Strategic alignment and value maximization for IT project portfolios

I. Robert Chiang · Manuel A. Nunez

Published online: 29 May 2012
© Springer Science+Business Media, LLC 2012

Abstract Managing project portfolios has been a challenge to many IT organizations due to the size and complexity of their initiatives that are often cross-functional, fast changing, and transformational in nature. A governance process on project solicitation, evaluation, and monitoring is thus essential to ensure the resulting portfolio creates tangible values, balances across priorities, and supports business objectives. An optimization model to streamline the decision processes for IT portfolios and programs is proposed. We consider project characteristics such as the extent of strategic alignment, expected benefit, development cost, and cross-project synergy to maximize the portfolio value. We also consider team proficiency and resource availability to determine a project portfolio that could be implemented within the overall development time. The multi-objective model identifies the optimal mix among project types and the solution procedure efficiently produces recommendations that are superior to those found with current empirical techniques. We also describe an evolutionary algorithm to find approximate solutions to the optimization model. Possible extensions on how the optimization procedure can go beyond projects to also streamline decisions such as the renewal or replacement of in-flight applications is discussed.

I. R. Chiang (✉)
Gabelli School of Business, Fordham University,
441 E. Fordham Road, Bronx, NY 10458, USA
e-mail: ichiang@fordham.edu

M. A. Nunez
School of Business, University of Connecticut,
2100 Hillside Road Unit 1041, Storrs, CT 06269-1041, USA
e-mail: mnunez@business.uconn.edu

Keywords IT project management · Project portfolio management · Strategic alignment · Project synergy · Multi-objective optimization · Evolutionary algorithms

1 Introduction

The IT organization of many enterprises is operating in a world of duality: supporting routine activities in business transactions, communication, and decision-making *while* creating game-changing capabilities in support of strategic goals and operation plans. This unique characteristic often leads to a heightened scrutiny and contention on IT resource allocation and investment selection due to the lasting, enterprise-wide implications of such decisions. Since routine operations (for technical infrastructure, in-flight applications, help desk, and so forth) often take up a significant proportion of the IT budget, maximizing the return from discretionary projects and initiatives has drawn significant interests from academics and practitioners alike.

Pervasive project escalations and costly operations have spurred several governance frameworks on system acquisition, development, operation, and management, including: PMI's PMBOK, ITIL, CobiT, and SEI's CMMI. These frameworks aim to achieving better productivity, quality, control, and compliance for application management, service delivery, and system development. Furthermore, a shift toward more incremental and agile software processes helps to improve the flexibility and progress visibility of software system development. However, focusing on the planning, execution, and management of applications or individual projects provides little assurance that, collectively, they constitute the best use of available resources or deliver the right set of products or services.

A holistic approach toward selecting and monitoring projects is thus essential to ensure the resulting project portfolio would align with the company's strategies, create business values, conform to available capacity and resources, and increase stakeholder satisfaction. Such approach is commonly referred to as project portfolio management (PPM). PMI defines PPM as "the centralized management of one or more portfolios, which includes identifying, prioritizing, authorizing, managing, and controlling project, programs, and other related work to achieve specific strategic business objectives." A program normally consists of several projects, other development works, ongoing operations, and IT-related assets. A program manager thus needs to balance resources demands across work types and report performance metrics while meeting operating goals. The value proposition of PPM (see [34], p. 4) includes its enablement on:

- investing in the right initiatives,
- optimizing available capacity,
- ensuring precise execution,
- creating acceptable level of changes, and
- realizing promised benefits.

Despite its obvious benefits, advancing PPM requires process improvement and sometimes a cultural shift that is by no means trivial. For example, without a combination of shared project roaster, communicated business imperatives, and documented evaluation criteria, it is difficult to determine the set of "right" initiatives with confidence. Similarly, unless there is continued risk assessment, progress monitoring, and status updates, an IT organization cannot easily gauge whether a project is on track to deliver benefits as promised. Shunning PPM altogether, however, could lead to projects' funding and resourcing being driven by hunches, decibel levels, and back-door channels (e.g., a denied request gets the funding anyway from an executive's "personal cache").

Several field studies [3, 23, 28] have observed that to achieve success in PPM there is a need to follow a gradual and disciplined path. The suggestion is to break down each portfolio review cycle into stages such as "define, select, execute, evaluate, evolve, and communicate" (as in [3]) for better process visibility and control. Gaining expertise in each stage gradually affords an organization both longer planning horizon (having clarity into both current and future projects) and shorter review cycles (the portfolio is balanced on a regular basis), allowing a project to go through multiple portfolio-level reviews during development. Other studies [23, 34] indicate that it has become commonplace in the current practice to have PPM-designated roles and teams to bolster visibility and accountability. Organizations do so by setting up a PPM Governance Council (PGC) in conjunction with an overarching program management office (PMO). The PGC, representing the side of business operations, consists of senior decision makers from key operating groups, corporate finance, and (strategic and tactical) planning functions (see [23], p. 55). The PGC communicates strategic directions, provides input in determining project contributions and priorities, and is empowered and entrusted on funding decisions. In particular, they also help in rating the degree of linkage between projects and strategy, and in determining the synergy among project proposals [23]. Therefore, having tools that increase process efficiency, reduce decision cycle time, and support PGC decision making can play a very important role in countering the sheer scope, span, and complexity that are commonly associated with portfolio-level decisions.

Among the promising areas in improving the IT PPM capability, our focus is on improving the project evaluation process and the decision support tools/technology. The first advancement in this paper is on allowing projects of different types to be evaluated under a single framework. The word portfolio implies that there are several classes of investments involved, and true to their all-encompassing roles, IT projects have been categorized as "infrastructure, utility, enhancement, and frontier," "run the business, grow the business and transform the business," or "utility, growth, and transformation" (see [3] and [23], p. 109). Because projects from different types/categories could vary significantly in size, risk level, strategic implications, and underlying technologies, they are often measured by a disparate set of metrics—even though they share the same pools of financial, personnel, and technical resources. For example, financial justifications on ROI, IRR, and NPV are common when ranking utility and enhancement projects; scoring sessions are then added for frontier projects to assess strategic contributions. On the other hand, real option analysis helps to justify infrastructure investment projects. Segregating portfolio decisions by category, however, prevents a true portfolio-level evaluation for missing out the *cross-category* intricacy commonly observed among software systems. Our unified model can simultaneously examine the collective financial, strategic, and architectural merits from every project under consideration.

Another process-related contribution of the current work is that it considers *cross-project* interactions. Prior works [16, 25] on project selection have often assumed that projects are "orthogonal" of each other, thus treating their joint value as additive. In reality, it is not uncommon to find either similar sets of functionality in different proposals or complementary features with add-on benefits than originally estimated. The pair-wise interrelationship between projects can sway the resulting portfolio value either positively (e.g., building business intelligence

capability boosts the utility of a data warehouse initiative) or negatively (e.g., same functionality, differ only in underlying technology). This in turn may produce incentive to bundle projects or to make them mutually exclusive of each other. Notice common scoring and ranking techniques, such as cost-benefit analysis or risk-reward plot, do not consider such interrelationship and can overstate (or understate) the value of their portfolio recommendations.

The third contribution of this work is the efficiency of the solution procedure. Accelerating and improving the quality of the portfolio decisions is critical, especially when the number of proposals and on-going projects could number in the thousands (see [32], p. 124; and [23], p. 442). Even for a problem of more moderate size, it is unrealistic to expect the PGC to make relevant decisions in a timely manner without proper decision and computational support. By comparing results from exhaustive search and evolutionary heuristics based on the notion of Pareto efficiency, we show that we can significantly reduce the computation time for portfolio evaluation while improving the consistency of the solution quality.

We summarize here several main results from this study. First, by enumeration we are able to confirm that, as business priority shifts, the optimal portfolio can change significantly even when the problem size is very small. We also show that using common metrics such as ROI does not accurately predict a project's inclusion and that a very compelling project could still be rejected because the level of resulting change is beyond what the organization can absorb. Second, by using the notion of Pareto optimality, our approach identifies a set of high-quality candidate portfolios, which is often preferred over a singular recommendation as the starting point for PGC deliberation. Comparing against enumeration results we can verify that our evolutionary algorithm quickly identifies the majority of portfolios on the efficient frontier. Third, the characteristics of our evolutionary approach indicate that our solution procedure can easily support decisions of much larger scale without suffering from exponential increase on computation time.

The rest of the paper is organized as follows: in Sect. 2 we review related research on IT portfolio, program, and project management. Section 3 presents the PPM process swimlane. Section 4 discusses our optimization model. In Sect. 5 we discuss practical issues in implementing our optimization mode. Section 6 is divided into two parts: Sect. 6.1 uses several numerical scenarios to demonstrate model behavior and solution optimality; and Sect. 6.2 introduces an efficient evolutionary algorithm to find approximations to the Pareto-optimal set. Finally, Sect. 7 summarizes our research and suggests potential directions for future study.

2 Literature review

Project portfolio management and financial portfolio management are similar in that they are concerned with value maximization and risk management; the IT field is known to leverage many best practices from the financial markets [26]. However, there are also significant differences that make the management practices and solution procedures not easily transferrable. For example, software projects encode intellectual properties to create one-of-a-kind systems that are tailored to a particular enterprise, a functional unit, or even certain individuals. There is no exchange or resell market for product salvage; nor there is reliable, standardized, or publicly available historical data to support portfolio decisions. In addition, the complexity of financial portfolio management is concentrated on the upfront analysis and selection; IT projects, on the other hand, only yield positive results if sound selection decisions are followed by precise execution within each of the lifecycle phases. PPM thus requires a high degree of coordination among portfolio, program, and project management while dealing with complex technical infrastructure and business mandates.

The core of IT PPM is project selection and resource allocation, which has been a research stream that dates back for nearly half of a century (e.g., see early works of [1, 21]). The solution approaches for this class of problem range from scoring methods that rank projects against a set of criteria [29] to more quantitatively-driven methods that involve optimization modeling [19] and dynamic programming [18]. Specific solution techniques cited in recent literature are numerous, including: Data Envelopment Analysis [15, 25] that use the efficient frontier concept to support both subjective and objective evaluations; decision rules and expert systems [7]; mathematical programming [16, 19]; simulation [9]; search heuristics such as ant colony optimization and genetic algorithms [13, 27]; real options [5, 6], among others. Even though the term "portfolio" appeared in several of the aforementioned papers, the majority of them essentially assume the projects under evaluation are of either R&D or of new product development nature. Such homogeneity does not represent IT project portfolios due to the presence of other project types such as application enhancements and strategic/infrastructure build-outs. Essentially, much of the previous work has missing terms in the objective function to support IT PPM.

In this paper, we adopt the concept of evolutionary algorithms (EA) in multi-objective optimization [11] as the solution technique. The use of EAs is an attractive approach for IT project selection because, at the early stage of a project lifecycle, decision makers are more interested in scanning high-quality candidates for the one that "meets

the eyes" than trying to identify the project portfolio that yields the highest objective value. Our EA procedure not only identifies candidate solutions that exhibit Pareto optimality, but also suggests how projects should be assigned to development programs so that the overall development time span can be reduced.

Another research stream leading to IT PPM is the corporate portfolio management (CPM) framework in general and PPM studies (see [23, 34]) in particular. CPM, as an approach for strategic planning, has a simple premise: to align discretionary investment expenditures with the organization's financial, strategic, and risk objectives (see [32], p. 1) in order to optimize resource allocation and maximize value creation. PPM instills CPM, both on process discipline and organization behavior, into a project setting to ensure that projects and programs fulfill organizational goals, strategies, and objectives. PPM also provides a holistic and systematic approach to project management by open communication and fact-based decision making.

The PPM decision criteria being applied to project proposals and business cases can be categorized [34] into four distinct groups: value, strategy, balance, and linkage. Besides the obvious needs to demonstrate (tangible) value and strategic alignment, the "balance" dimension helps ensure that the portfolio mixes projects with varying degrees of risks, makes sensible distribution of available resources among functional areas, and produces both immediate capability as well as sound foundation for the future. There is often a high level of uncertainty [24] associated with IT projects due to their innovative and complex nature and the fact that selection decisions are often made very early during the project lifecycle. Frameworks on diagnosing, clustering, and mitigating project-level risks [2, 31, 35] have been supplemented by those dealing with program-level risks [30] and organization-level risks [10]. While much of the work cited earlier in the section can be applied toward the resource allocation problem, the "now and future" tradeoff has received less attention. Different from the real option analysis, we propose a more tractable approach so that future capabilities can be used to influence the selection of current projects even though they do not require immediate attention with respect to selection, allocation, and assignment.

As for the "linkage" dimension of PPM decisions, due to project interdependency and resource sharing, selection decisions made in isolation based on individual project's merit likely leads to suboptimal results [15]. Prior work [14] has treated such interaction in three categories: resource, benefit, and outcome. We consider such interactions in our model when projects are shown to have complementary effects (a cluster of projects should be funded simultaneously) or substitutable effects (when one among several should be chosen).

3 Project portfolio management

The PPM cycle can be illustrated by the swimlane shown in Figure 1. First, project proposals are solicited from functional areas, levels of management, end users, and within the IT organization. Following an initial screening, promising proposals are elaborated into a business case if necessary. A business case contains additional sections such as as-is business landscape, justification/value propositions, financial benefits, cost implications, risk assessment, technical alternatives, and known dependencies. The EMPO then verifies project attributes (by PMO analysts), assesses strategic contribution, and confirms interrelation among projects (jointly by program managers and business architects, with input from PGC).

For each project category, the PGC determines the decision time horizon, weighing of project (strategic, financial, and architectural) attributes, acceptable level of changes, and levels of resources. Once the value, risk, interrelationship, and resource parameters are set up, the optimization model is solved to suggest which projects to fund and to which development program each project should be assigned. The model solutions are derived without exceeding the budgetary, capacity, or risk levels. An unfunded proposal/business case could receive no further consideration or be placed in the list of "future projects" for upcoming rounds. Approved projects go through periodic reviews to ensure relevance and proper progress toward completion; the ones that no longer demonstrate sufficient merit could face termination. Completed projects are moved under the realm of application portfolio, and capacities allocated to them will be reclaimed back to the resource pool.

It is pointed out that the steps shown in shaded boxes in Fig. 1 are those to which we have made unique contributions. For example, process steps five, six, and seven collectively ensure that projects can be compared against the same set of criteria regardless of type; process step eight is the optimization engine that can efficiently identify a set of candidate portfolios; it also performs quick sensitivity and what-if analysis when there are subjective or external mandates (e.g., green-lighting a project due to regulatory compliance) from the decision makers.

We are assuming each program handles its own value management, project scheduling, resource allocation, and stakeholder relationship. Therefore, these activities are considered out of scope at the portfolio level. The main concern here is to ensure that the workload is balanced among programs and that each project is assigned to a program with suitable skill set and resource availability, as shown in process step nine. The EPMO consolidates program status and performance reports for PGC and facilitates inter-program coordination on risks, issues, and resource allocation.

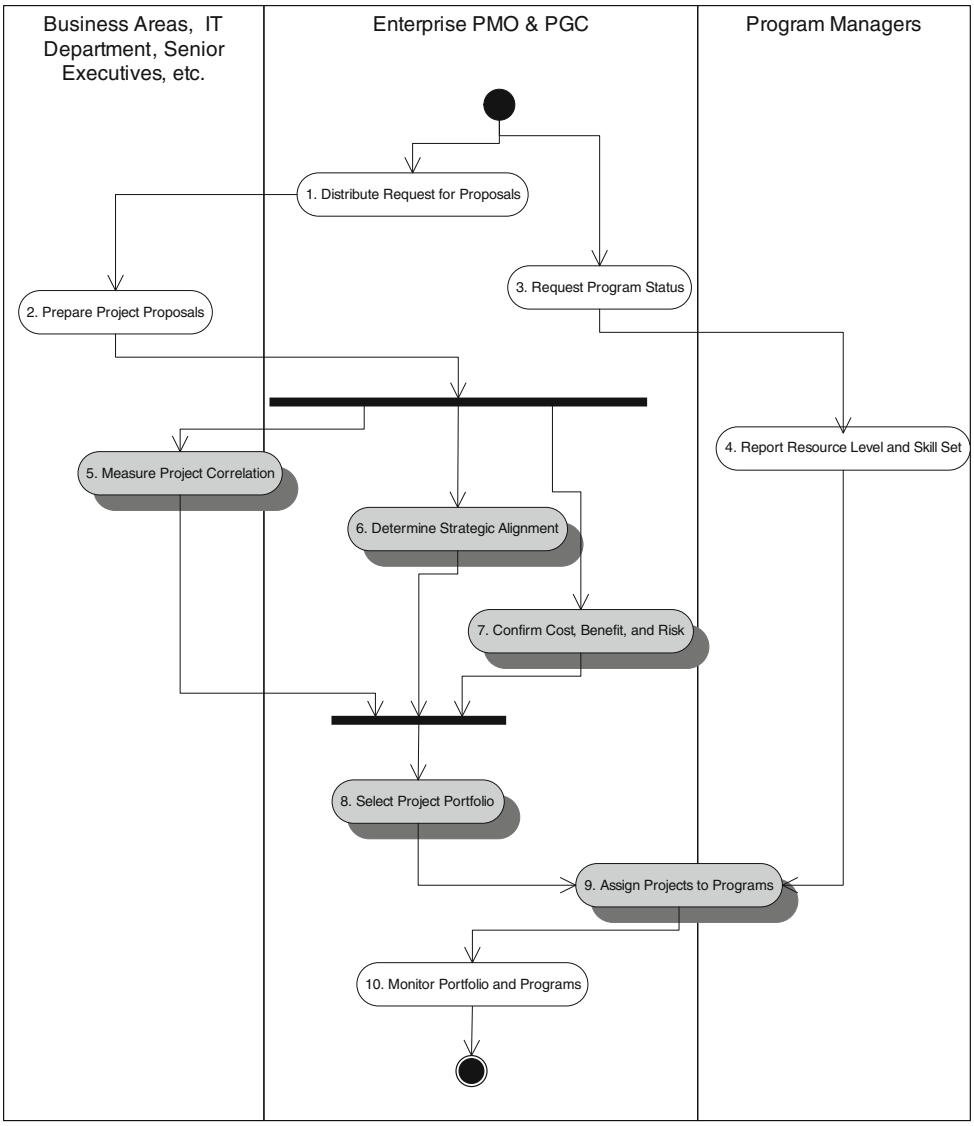

Fig. 1 PPM process swimlane

4 Project portfolio optimization model

4.1 Overview

We consider a set of project proposals from which a company has to select a subset of proposals into a portfolio of projects to be implemented. The selection will be based on multiple objectives: the overall financial benefit of the selected proposals (if successfully completed), the strategic alignment of the portfolio, and the infrastructure contribution to future projects. In addition, the portfolio has to satisfy a set of constraints, namely, its overall cost should not exceed a given budget amount, its overall risk should be below a maximum level of tolerated risk, it should support strategic objectives, and the duration of the projects in the portfolio should not exceed a target time duration. To control the time duration of the projects in the portfolio, we assume that the company has available teams of resources that can be assigned to projects. Hence, an optimal proposal selection should also include a resource assignment plan.

In formulating a mathematical model for this project portfolio problem, we first consider the selection of proposals to be in the portfolio. The notation for the model is given in Table 1. In particular, let M denote an index set of project proposals under review. For each $i \in M$, we denote by x_i the binary decision variable representing whether proposal i will be included in the portfolio ($x_i = 1$) or not ($x_i = 0$). Thus, the composition of each portfolio is fully determined by an M-dimensional 0–1 vector of x_i values, and it is easy to see that there could be up to $2^{|M|}$ feasible portfolios in the worst case. In light of this, we will frequently use X to denote an M-dimensional 0–1 vector and to represent a project portfolio at the same time.

We next discuss the formulation of the objective functions and constraints in terms of the decision variables x_i.

Table 1 Notation summary for the project portfolio selection problem

Sets and indexes	
i, l	Indexes for current project proposals
i'	Index for future project proposals
j	Index for strategies
k	Index for objectives
p	Index for programs
M	Index set of current project proposals
M'	Index set of future project proposals
N	Index set of strategies
O_j	Index set of objectives for strategy j
P	Index set of programs
Parameters	
r_i	Return of project i if successful, $i \in M \cup M'$
u_i	Probability of failure for project $i, i \in M \cup M'$
ra_i	Risk adjusted return for project $i, i \in M \cup M'$
c_i	Cost of project $i, i \in M$
$d_{ii'}$	Dependency of project i' on project $i, i \in M, i' \in M'$
f_{il}	Functional overlap between project i and project $l, i, l \in M$
s_{ijk}	Contribution of project i to objective k of strategy $j, i \in M, k \in O_j, j \in N$
w_j	Weighted importance of strategy $j, j \in N$
g_p	Resource-hours available to program p per time unit, $p \in P$
h_{ip}	Resource-hours required to complete project i if assigned to program $p, i \in M, p \in P$
B	Available budget
U	Risk tolerance
T	Target time to complete all projects
Decision variables	
x_i	Whether or not project i is implemented, $x_i \in \{0, 1\}, i \in M$
y_{ip}	Whether or not project i is assigned to program $p, y_{ip} \in \{0, 1\}, i \in M, p \in P$
t_p	Time required to complete all projects assigned to program $p, p \in P$
X	Portfolio of projects to be implemented as a vector of decision variables
Objective values	
α	Relative importance of strategic alignment
β	Relative importance of financial benefit
AA(X)	Infrastructure dependencies for portfolio X
SA(X)	Strategic alignment for portfolio X
AFB(X)	Adjusted expected financial benefit for portfolio X
CC(X)	Convex combination of SA, AFB, and AA objectives for portfolio X
z	Maximum convex combination

4.2 Objective functions

We first model the financial value/benefit aspect of a project. Most projects are expected to generate tangible benefits such as revenue increase, cost reduction, and production/service quality improvement. The stated benefit projection should be adjusted by the likelihood of a project's technical failure (i.e. not fully developed and deployed) and commercial failure (i.e. a system's total cost of ownership outweighs realized gains). One example approach in previous studies has been to use the risk level as a part of the discount factor when computing ROI and NPV; another approach is to show both types of failure as possible outcomes on a benefit decision tree [23]. To account for this risk, let r_i be the full benefit estimation for project i if successfully completed and let u_i $(0 < u_i < 1)$ be the project's probability of failure, then the expected benefit of project i is:

$$ra_i := 0 \times u_i + r_i \times (1 - u_i) = r_i(1 - u_i). \quad (1)$$

Notice that we assume that if the project fails the benefit from the project will not be realized. Expression (1) is a

risk-adjusted estimation of the benefit from a project and it is a common performance measure used in practice [23].

Incorporating decision variables, the expected financial benefit of a portfolio is given by

$$\mathrm{EFB}(X) := \sum_{i \in M} ra_i x_i.$$

However, this term does not account for the pair-wise correlation between projects. When two projects complement each other, their implementation can lead to a positive synergy that may lead to financial benefits beyond just the sum of the two projects' individual risk-adjusted benefit. For example, creating customer master databases for both commercial and residential lines of a business may create cross-selling opportunities at corporate level. On the other hand, having two projects with significant functional overlap may lead to a cost increase that surpasses the combined benefit. Hence, pairs that exhibit a positive correlation could be favored jointly in the portfolio, whereas pairs of functionally overlapped projects should be avoided. To model the impact of proposals correlations we use parameter f_{il} representing the correlation between proposals i and l, where $f_{il} = f_{li}$. As expected, the sign of f_{il} helps to describe whether two projects are synergetic (>0), neutral (=0), or have a cannibalizing (<0) relationship. We then adjust the expected financial benefit as follows

$$\mathrm{AFB}(X) := \sum_{i \in M} ra_i x_i + \sum_{i,l \in M} (ra_i + ra_l) f_{il} x_i x_l, \tag{2}$$

where the first term of AFB adds the contribution of individual projects whereas the second term captures the effect of pair-wise correlation between projects. Notice that for a particular pair (i, l) of proposals included in the portfolio ($x_i = x_l = 1$), the contribution of the pair to AFB is

$$\begin{aligned} & ra_i x_i + ra_l x_l + 2 f_{il}(ra_i + ra_l) x_i x_l \\ & \quad = (1 + 2 f_{il}) ra_i + (1 + 2 f_{il}) ra_l, \end{aligned}$$

so that the individual expected benefits of the two projects are proportionally magnified or reduced depending on the sign of f_{il}. The value of f_{il} is influenced by project factors such as the cost difference between developing related project simultaneously versus independently, or business factors such as the extent of cannibalization when funding similar applications within a market segment. Notice that because a project's inclusion or exclusion to a portfolio could potentially affect the valuation of many others due to the ripple effect shown in the second term of (2), it is no longer possible to just rank projects based on its own merit and select highly-ranked projects until a cutoff is reached (e.g., sum of project costs exceeds the budget.)

Besides financial motives, another key dimension of portfolio evaluation is how software projects help realizing an organization's visions, goals, and strategies. Each project's contribution needs to be evaluated both for necessity and sufficiency in terms of the level of support that each strategy demands. Naturally, projects with strong strategic alignment should be given higher consideration. To ensure sufficiency of support, projects that are critical to the success of a strategy may be chosen even if they have less attractive profile on the financial/benefit side. To better quantify strategic alignment (SA), one approach is for the EPMO to break down each strategy into a set of high level objectives. Working with senior executives and business architects, the PMO analysts can then assess each project's strategic implication by matching its features into such objectives for the degree of contribution. This process is akin to the Ends-Means-Projects mapping detailed in [23]. To ensure higher-priority strategies receive better project support, techniques such as AHP (analytic hierarchy process), Delphi method, and other consensus-building mechanisms have been widely used to rank strategies by numerical scales rather than just an ordinal sequence. This ranking of strategies is modeled by assigning non-negative weights w_j to each strategy and normalizing so that

$$\sum_{j \in N} w_j = 1.$$

This way we can capture the importance of the strategies relative to each other. The degree of support that project proposal i provides to goal k from strategy j is denoted by $s_{ijk} \geq 1$; the higher this value the more aligned the project to a given goal. By combining the strategy weights (ranking) with the proposals' support values we obtain the overall portfolio's ability to support the strategic initiatives as:

$$\mathrm{SA}(X) := \sum_{i \in M} \sum_{j \in N} \sum_{k \in O_j} s_{ijk} w_j x_i. \tag{3}$$

In particular, the overall strategic contribution of proposal i is

$$\sum_{j \in N} \sum_{k \in O_j} s_{ijk} w_j,$$

which averages the degree of support of this proposal to all the company's goals weighted by the relative importance of each strategy. Projects that align with high-priority initiatives and support more objectives within such initiatives are more valuable than otherwise. As a quick double-check for inter-project interaction, projects based on similar technical architecture but cover different sets of objectives have higher potential for synergy; two projects fulfilling the same set of strategic objectives could decrease portfolio return by preempting other promising initiatives from getting funded.

The final dimension of portfolio evaluation is to ensure the portfolio is balanced. The concept is akin to the balanced scorecard in the strategic planning literature, in which resources should be allocated not only for near-term financial gains and process improvements but also to create a level flexibility both on the technology front and at the enterprise level. Such futuristic view is essential to enabling future strategies and projects. To capture the contribution to architectural agility/flexibility from such (organizational and technical) infrastructural investments, we consider parameter $d_{ii'} \in \{0, 1\}$ representing whether there is a causality dependency between projects i and i' ($d_{ii'} = 1$) or not ($d_{ii'} = 0$). In other words, $d_{ii'} = 1$ means that to implement project i' it is required to first successfully complete project i. Therefore, the overall value of planned projects that have a dependency on the current portfolio's infrastructure is given by

$$\mathrm{AA}(X) := \sum_{i \in M} \sum_{i' \in M'} (1 - u_i) d_{ii'} ra_{i'} x_i, \tag{4}$$

where we again multiply by $1-u_i$ to account for the successful completion of project i in the portfolio. Notice that if proposal i is selected to be in the portfolio and it is successfully completed (with probability $1-u_i$), it will enable all the future projects i' with $d_{ii'} = 1$, which subsequently will yield an expected benefit of $\sum_{i' \in M'} d_{ii'} ra_{i'}$. Since infrastructure investment is normally hard to justify from the financial perspective, many IT organizations simply resort to placing a "License to Operate" label to approve such investments. With proper due diligence on the "dependency matrix" among projects, however, it is possible to estimate the extent of capability/ flexibility such projects provide. The AA measure shown in (4) is different from the interrelationship term in AFB, even though both are designed to capture inter-project dependency. A project's AA contribution is directional, stemming from the foundation project to the applications relying on it; the interrelation AFB is bi-directional (mutual) and most likely exists among applications.

4.3 Constraints formulation

A realistic view into the various constraints is also centrally important to a portfolio's success because doing so increases the chance of delivering results as promised without resorting to heroics, burnouts, or mere coincidence. We consider four sources of restrictions: budget, risk tolerance, support sufficiency of objectives, and resources. For the budget constraint, we use a traditional approach in which each selected project incurs a sunk-cost c_i and the total cost of all projects in the portfolio should not exceed a given amount B. This yields the constraint

$$\sum_{i \in M} c_i x_i \le B.$$

For the risk-tolerance constraint, the portfolio's risk score is a weighted average of the probabilities of failure u_i of the projects in the portfolio, where the weights are based on project cost/overall cost, that is, the weight for proposal i is given by

$$\frac{c_i}{\sum_{i \in M} c_i x_i}.$$

Hence, we want the risk score not to exceed the risk tolerance threshold of U:

$$\sum_{i \in M} \left(\frac{c_i}{\sum_{i \in M} c_i x_i} \right) u_i x_i \le U.$$

This way, proposals with relatively low cost ($c_i / \sum_{i \in M} c_i x_i$ low) but risky (u_i high) still have a chance to be considered into the portfolio as long as they provide a high-reward (aggressive projects); a costly high-risk, high-reward project would pass the crucible only if the risk threshold is sufficiently high to accept the possibility of its outright failure. Notice the level of U is a proxy for the enterprise's tolerance toward risk, which is influenced both by the development organization's ability to deliver results and how much the business units can absorb new capabilities brought forth by the projects.

Sufficiency of support for strategic objectives means that each objective of each strategy should be covered by projects features in the portfolio:

$$\sum_{i \in M} s_{ijk} x_i \ge 1,$$

for all $j \in N$ and $k \in O_j$.

The resource constraint is expressed in terms of the time it would take to complete all of the projects in the portfolio. Typically a company has a set of *programs* to which projects in the portfolio are assigned. Each program consists of a group of resources (staff, equipment, etc.) dedicated to the projects assigned to the program. A program's personnel capacity during a time period of t_p time units is given by $g_p\, t_p$, where g_p is the total number of hours that all the program members work per time unit. For example, if the program has ten staff members who work 8 h per day, and the program runs for a period of thirty working days, then $g_p = 80$ h/day and the capacity of the program is $g_p\, t_p = 2{,}400$ person-hours during the time period. On the other hand, if a project i requires h_{ip} person-hours to be completed by program p, then we want $h_{ip} \le g_p t_p$. Hence, when considering all projects and all programs, we obtain the following constraints:

$$\sum_{i \in M} h_{ip} y_{ip} \le g_p t_p,$$

for all $p \in P$, where y_{ip} is a binary variable indicating whether or not project i has been assigned to program p. Since each project must be assigned to exactly one program, we introduce the constraint

$$\sum_{p \in P} y_{ip} = x_i,$$

for all $i \in M$. Finally, to ensure that the selected projects are completed within the planning horizon by each program, we add the constraint $t_p \leq T$, for all $p \in P$. Note that without loss of generality we have assumed that the number of resources-hours available to program p is a fixed value g_p. This is because the optimal assignment of projects to programs can be easily re-computed when there are changes in available resources or constraints. In fact, a sensible decision-maker would study several scenarios and perform sensitivity analysis by varying these parameters before reaching a final plan.

4.4 Optimization model

We can now combine the three evaluation dimensions with the four sources of restrictions to obtain the following optimization problem:

$$\text{(PS)} \quad z = \max \text{CC(X)} = \alpha \text{SA}(X) + \beta \text{AFB}(X) + (1 - \alpha - \beta)\text{AA}(X),$$

$$\text{s.t.} \tag{5}$$

$$\sum_{i \in M} c_i x_i \leq B,$$

$$\sum_{i \in M} u_i c_i x_i - U \sum_{i \in M} c_i x_i \leq 0, \tag{6}$$

$$\sum_{i \in M} s_{ijk} x_i \geq 1, \forall j \in N \text{ and } k \in O_j, \tag{7}$$

$$t_p \leq T, \forall p \in P, \tag{8}$$

$$\sum_{i \in M} h_{ip} y_{ip} \leq g_p t_p, \forall p \in P, \tag{9}$$

$$\sum_{p \in P} y_{ip} = x_i, \forall i \in M, \tag{10}$$

$$x_i \in \{0, 1\}, y_{ip} \in \{0, 1\}, t_p \geq 0, \forall i \in M \text{ and } p \in P.$$

The goal is to maximize the convex combination of the strategic alignment, financial benefit, and portfolio balance objectives. Naturally, when considering utility (routine updates and bug fixes) projects, more weight should be given to financial considerations. Major enhancement or new capability creation projects should also be evaluated on the merit of strategic alignment, while transformational and infrastructure projects should be judged mainly for its ability to provide flexibility and alignment for the future of the organization. Constraint (5) indicates that the overall budget has to cover the cost of individual projects. Constraint (6) is to ensure the risk level of the portfolio does not exceed a threshold level. Constraint (7) ensures that every objective of each strategy has its critical projects included in the portfolio. Constraint (8) indicates that the time to complete the projects assigned to a program should not exceed the target time for the completion of all projects, that is, $T \geq \max\{t_p : p \in P\}$. Constraint (9) indicates that the total time spent in each of the projects assigned to a program should not exceed the total time available to the program. Finally, constraint (10) indicates that exactly one program must be chosen for project i if the project has been selected for implementation ($x_i = 1$) and no programs should be chosen for project i if not selected ($x_i = 0$).

In our model we have assumed that project cost c_i stays relatively the same regardless of the efficiency of the team assigned to it. This would be the case in situations where $c_i = h_{ip}\rho_{ip} + \delta_i$ for all $p \in P$, where ρ_{ip} is the cost per hour when team p is assigned to project i, and δ_i represents the cost of additional technical infrastructure components in project i. It means that the team efficiency (as reflected by a lower h_{ip} value) is offset by the personnel cost due to a more experienced and likely pricier staff members (ρ_{ip} is high). Nevertheless, if this is not the case, it is not difficult to modify the model by replacing constraint (5) by the following constraint

$$\sum_{i \in M} \sum_{p \in P} \left(h_{ip}\rho_{ip} + \delta_i\right) y_{ip} \leq B,$$

to account for the efficiency of the teams.

Also notice that even though ra_i is positive, it is possible that the net expected profit, $ra_i - c_i$, is negative in an optimal solution. This does not represent a contradiction in our formulation because it is perfectly reasonable to have a project with negative net profit in an optimal portfolio as long as it maximizes the two other objective functions we consider: strategic alignment and infrastructure. For instance, a project that enables the implementation of many low-risk high-yield future projects could be very desirable even if its implementation represents a temporary loss for the company.

Finally, as for the complexity of solving problem PS, notice that problem PS is a quadratic assignment problem and since quadratic assignment problems are well-known to be *NP*-Hard, it is likely that there is no efficient (polynomial) algorithm to solve it.

5 Model implementation in practice

As indicated in Table 1, the formulation of a data instance of problem PS requires the determination or estimation of

several parameters. These parameters can be grouped into three categories:

1. Financial Parameters: including the return on the projects r_i, the probabilities of failure u_i, project costs c_i, available budget B, and risk tolerance U.
2. Strategic-Infrastructure Parameters: including dependencies $d_{ii'}$ and functional overlaps f_{il} among projects, their strategic contribution s_{ijk}, and the weighted importance of the strategies w_j.
3. Resource Parameters: including resource-hours required per project h_{ip}, resource-hours available to programs g_p, and the target completion time for the projects in the portfolio T.

The practical considerations on how to determine the financial parameters is a well-studied subject that has received considerable attention in the literature of project portfolio management and so, we do not further discuss these parameters. Levine [23] for example has an extensive discussion on this topic. Similarly, the estimation of the resource parameters is amply studied in the literature of project management (see for instance [33]). Therefore, we concentrate on discussing the strategic-infrastructure parameters.

The dependencies of future project proposals on current project proposals $d_{ii'}$ are determined by the technical requirements of the future projects and/or resource requirements of the future projects that will be made available by implementing current projects. For example, a current project consisting of performing a server migration would make possible at a later stage to have a platform for implementing other projects such as the installation of a wide range of different business intelligence software tools (reporting, dash boards, ad hoc queries, etc.) that cannot be implemented because the tools are not compatible with the current architecture or because the current equipment has become obsolete. This type of dependencies can be determined by consulting the engineers or other technical support personnel related to the project. For example, in the server migration project the infrastructure services team would determine the feasibility and potential dependencies of future projects after the migration.

Functional overlaps f_{il}, strategic contributions s_{ijk}, and strategy importance weights w_j can be determined by using a ranking scoreboard approach. For example, for the strategic contributions a score matrix is created where each row represents a project i and each column represents a strategic objective (j, k). Those who make the ultimate decisions regarding which projects will enter the portfolio, technical consultants, managers of affected departments, etc. would complete the matrix assigning a numerical value to the degree of linkage between projects and strategic objectives as perceived by each participant. Then the value of s_{ijk} could be the average for entry (i, j, k) across participants normalized to be within certain range (e.g., between 0 and 1). This technique is commonly used by companies with very successful results as indicated in [23, 34]. In fact, decision-support software tools have been created to streamline the process (e.g., Expert Choice software [23]). Similar techniques can be used to estimate functional overlaps, where a project-to-project matrix can be used and the participants would enter their perceived degree of overlap or complementarity for each pair of projects.

A final issue we wish to discuss is the choice of the weight that each of the objective functions (strategic alignment, financial benefit, and infrastructure) should receive in solving the convex combination objective function of PS. These weights are represented by α for the strategic alignment, β for the financial benefit, and $1 - \alpha - \beta$ for the infrastructure. Of course, we can once again use a ranking scoreboard approach as described above to determine the values of these weights. However, a more robust approach and perhaps more useful in decision making is to determine a *Pareto-optimal* solution set. For each feasible decision vector (feasible portfolio) $X \in \{0, 1\}^{|M|}$ corresponds an objective vector $F(X) \in \Re^3$ consisting of the evaluation of each of the three objective functions at X. Hence, a data instance of problem PS determines two spaces: the decision space (where X lives) and the multi-objective space (where $F(X)$ lives). A vector v in the objective space is said to dominate another vector w if $v_i \geq w_i$ for $i = 1, 2, 3$ and $v_i > w_i$ for at least one index i. Therefore, the Pareto-optimal set associated with problem PS is defined as the set of feasible objective vectors that are not dominated by any other feasible objective vector.

An easy to prove result is that for a given α and β, the multi-objective vector associated with an optimal solution of problem PS has to be in the Pareto-optimal set associated with PS. Therefore, if we first compute the Pareto-optimal set of a data instance of problem PS, then the solution of PS can be found by maximizing $CC(X)$ across the Pareto-optimal set. Moreover, by knowing all the Pareto-optimal solutions, the decision makers would be at a better position to study all the tradeoffs between the three objective functions. In the next section we describe an efficient heuristic to determine approximate Pareto-optimal solutions.

6 Numerical experiments

6.1 Scenario analysis

We consider several scenarios to illustrate the interaction of the different optimization factors in play within our model formulation. We start analyzing the effect of return and risk on the optimal solution in a situation where the

Table 2 Scenario 1 parameters

Project i	Return r_i	Risk u_i	Cost c_i	Return/cost
1	88	0.49	55	1.60
2	90	0.27	10	9.00
3	93	0.21	91	1.02
4	97	0.05	74	1.31
5	88	0.29	77	1.14
6	98	0.33	25	3.92
7	92	0.79	60	1.53
8	64	0.63	22	2.91

strategic alignment is the same for all projects under review. Consider a portfolio selection problem with eight project proposals, two equally weighted strategic goals and two objectives per strategy (for a total of four objectives). We also assume that there are two programs to complete the selected project proposals and four future projects that might or not depend on the selected proposals. Table 2 summarizes the parameters of the problem instance.

In this scenario the strategic contribution of each project to each of the four objectives is the same ($s_{ijk} = 1$) and the functional overlap values are zero ($f_{il} = 0$). The risk tolerance is $U = 0.5$ and $T = 15$ (so that corresponding constraints are not binding). We find the Pareto-optimal frontier of problem PS as the budget varies from 50 to 450. Table 3 shows the results; the expected return column is based on the highest return obtained across Pareto-optimal solutions for each budget instance (i.e. β dominates α and $1 - \alpha - \beta$).

Notice that as the budget increases more projects are selected and the selection is driven by the ratio of the return to the cost as expected. Projects 2 and 6 with the highest ratio are always chosen.

We next consider a different scenario in which we study the effect of strategic alignment. In the second scenario we assume that the returns, risks, and costs are the same for all the projects, but the contributions to strategic goals s_{ijk} change as indicated in Table 4. Concretely, we assume that all the projects have the same return r_i of 50, same risk u_i of 0.2, and same cost c_i of 20. We also set the weights of the two strategic goals as $w_1 = 0.2$ and $w_2 = 0.8$. All the remaining parameters remain the same as in the first scenario.

Table 5 shows the highest strategic contribution across Pareto-optimal solutions (i.e. α dominates β and $1 - \alpha - \beta$) as the budget ranges from 20 to 100. Projects are greedily added to the portfolio in the sequence $7 \rightarrow 2 \rightarrow 1 \rightarrow 8 \rightarrow 3 \rightarrow 6 \rightarrow 4 \rightarrow 5$ according to their net contribution to strategic alignment. The first projects to be added as the budget increases are those who contribute to all objectives, then those who contribute more to strategic goal 2 (with the highest weight $w_2 > w_1$), and finally the projects who only contribute to strategic goal 1.

Table 3 Solutions as the budget increases in scenario 1

Budget	Projects 1	2	3	4	5	6	7	8	Expected return
50	no	yes	no	no	no	yes	no	no	$394
100	yes	yes	no	no	no	yes	no	no	$529
150	no	yes	no	yes	no	yes	no	yes	$742
200	no	yes	yes	yes	no	yes	no	no	$891
250	yes	yes	no	yes	yes	yes	no	no	$993
300	no	yes	yes	yes	yes	yes	no	yes	$1149
350	yes	yes	yes	yes	yes	yes	no	no	$1213
400	yes	yes	yes	yes	yes	yes	no	yes	$1284
450	yes	yes	yes	yes	yes	yes	yes	yes	$1342

Table 4 Scenario 2 parameters

Project	Strategic goal 1		Strategic goal 2	
	Objective 1.1	Objective 1.2	Objective 2.1	Objective 2.2
1	1	1	1	1
2	1	1	1	1
3	0	1	1	0
4	0	1	0	0
5	1	0	0	0
6	1	1	0	0
7	1	1	1	1
8	1	0	1	1

Finally, we consider a scenario where the functional overlap among projects plays a significant role in the selection of projects. In this scenario we assume the same parameters r_i, u_i, and c_i as in the second scenario, except that we set the budget at $B = 40$, so that only two projects can be selected. Table 6 shows the functional overlaps among projects.

Table 7 shows two Pareto-optimal solutions in Scenario 3 corresponding to the highest expected financial benefit and the highest strategic alignment, respectively. Notice that the projects that contribute the most to strategic goals are not selected if we want to maximize expected financial benefit. this is because they have negative or zero overlap while the selected projects (projects 7 and 8) have positive overlap.

6.2 Evolutionary heuristic

Given the multi-objective nature of the project selection problem and that the complexity of solving data instances of problem PS increases with the number of project

Table 5 Solutions as the budget increases in scenario 2

Budget	Projects 1	2	3	4	5	6	7	8	Strategic contribution
20	no	no	no	no	no	no	yes	no	2.0
40	no	yes	no	no	no	no	yes	no	4.0
60	yes	yes	no	no	no	no	yes	no	6.0
80	yes	yes	no	no	no	no	yes	yes	7.8
100	yes	yes	yes	no	no	no	yes	yes	8.8
120	yes	yes	yes	no	no	yes	yes	yes	9.2
140	yes	yes	yes	yes	no	yes	yes	yes	9.4
160	yes	yes	yes	yes	yes	yes	yes	yes	9.6

Table 6 Scenario 3 parameters

Project	Projects 1	2	3	4	5	6	7	8
1	1	−1	0	0	0	1	0	0
2		1	0	0	0	−1	0	1
3			1	0	0	0	0	0
4				1	1	1	−1	−1
5					1	−1	0	0
6						1	0	1
7							1	1
8								1

proposals, it is necessary to develop fast heuristic methods to find approximations to the Pareto-optimal set associated with a data instance. One approach that has proven very efficient in finding good approximations is evolutionary algorithms (EA). In an EA the idea is to keep a fixed-size set of candidate solutions (portfolios) that moves toward the Pareto-optimal front of solutions of the PS as the number of iterations of the EA increases. At the same time, the EA maintains diversity within the set of candidate solutions by trading objectives (strategic alignment, financial benefits, and infrastructure) differently.

To implement an efficient EA we adapt a multi-objective genetic algorithm strategy described in [11]. In our algorithm, we first compute upper bounds on each of the three objectives under consideration, and then generate an "ideal" set of objective vectors by combining the upper bounds. For instance, if the upper bounds are $\mathcal{U}_1, \mathcal{U}_2$, and $\mathcal{U}_3$, respectively, then we generate three-dimensional objective vectors of the form $(\mathcal{U}_1, 0, 0), (0, \mathcal{U}_2, 0), (0, 0, \mathcal{U}_3)$, etc. We denote by $V^{**} \subset \Re^3$ the set of ideal objective vectors. Notice that ideal objective vectors might not have corresponding feasible solution vectors X in the space of decision variables. In other words, if $V^* \subset \Re^3$ denotes the Pareto-optimal set, then

Table 7 Pareto-optimal solutions in scenario 3

Projects 1	2	3	4	5	6	7	8	AFB	SA
no	no	no	no	no	no	yes	yes	400	3.8
no	yes	no	no	no	no	yes	no	240	4.0

it is possible to have $V^* \cap V^{**} = \emptyset$. On the other hand, creating V^{**} is computationally easy for a given data instance of PS.

Using the ideal set, we define the "fitness" of a decision vector X (chromosome) as

$$Fitness(X) := \frac{1}{1 + dist(F(X), V^{**})},$$

where $F : \{0,1\}^{|M|} \to \Re^3$ denotes the function that maps decision vectors into the three-dimensional objective vectors:

$$F(X) := \begin{cases} (\mathrm{SA}(X), \mathrm{AFB}(X), \mathrm{AA}(X)) & \text{if } X \text{ is feasible,} \\ (0,0,0) & \text{if } X \text{ is not feasible,} \end{cases}$$

and $dist(F(X), V^{**})$ denotes the Euclidean distance from the vector $F(X)$ to the set V^{**} in $\Re^3$. In other words, solutions close to the ideal set are rewarded as long as they are feasible. Infeasible solutions are penalized by mapping them to the origin in objective space, a point far away from the ideal set.

Diversity in each iteration of the EA is maintained by using the usual operators of genetic algorithms, that is, random selection according to fitness, crossover, and mutation, as applied to $|M|$-dimensional binary decision vectors. Furthermore, at iteration n the EA generates a set $W_n \subset \{0,1\}^{|M|}$ consisting of the decision vectors whose corresponding objective vectors are the dominant vectors discovered by the EA up to iteration n.

To update the set W_n from the set W_{n-1} at iteration n, we use the following algorithm: Let m denote the size of the population of chromosomes at iteration n. Denote by $X_1, \ldots, X_m$ the chromosomes (portfolios) in the current population, and $v_i := F(X_i)$, for $i = 1, \ldots, m$, the objective vectors associated with each of those chromosomes. Similarly, let V_{n-1} be the set of vector objectives associated with decision vectors in W_{n-1}. Set $W_n = \emptyset$ initially. For each $i = 1, \ldots, m$, if v_i is not dominated by the vectors in V_{n-1}, then add X_i to W_n ($W_n \leftarrow W_n \cup \{X_i\}$). Finally, for each $X \in W_{n-1}$ if $F(X)$ is not dominated by the current vectors in W_n, then add X to W_n ($W_n \leftarrow W_n \cup \{X\}$). The sequence of sets $W_1, W_2, \ldots$ is such that the objective vectors of the elements of W_{n+1} are the same or dominate the objective vectors of the elements of W_n. Hence, we

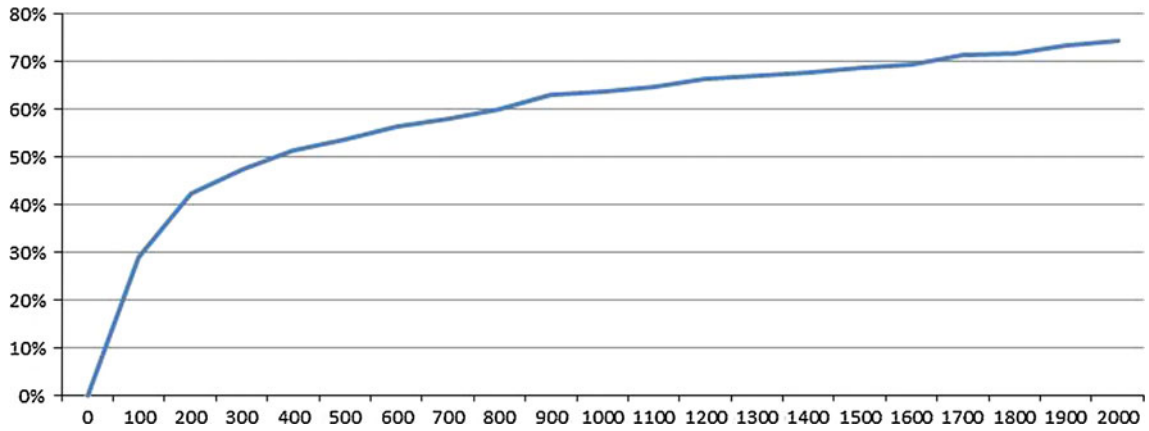

Fig. 2 Percentage of Pareto-optimal solutions discovered by EA as number of iterations increases

obtain a sequence of improving Pareto sets that converges to the Pareto-optimal front in the long run.

From a complexity point of view, the resulting EA has a running time that is polynomial in the number of project proposal, the chromosome population size, and the number of iterations of the EA. Hence, the EA is an efficient algorithm.

To test the EA, we generated a random sample of ten data instances with twenty project proposals each. We chose twenty project proposals because the data instance size is small enough that we can still use exhaustive search to find the Pareto-optimal set for each data instance in reasonable time, while at the same time having a large enough decision space so as to make the problem non-trivial to solve. We computed the Pareto-optimal sets for each data instance in order to have a benchmark to compare the performance of the EA. Notice that there could be up to $2^{20} \approx 1.05$ million feasible solutions in the decision space. In addition, each data instance has up to 25 strategic goals, ten future project proposals that depend on the current twenty proposals, and five programs to implement the selected projects. We ran the EA for 2000 iterations, each iteration consisting of a population of twenty chromosomes. This means that the EA explored at most 40,000 solutions, that is, at most 4 % of the decision space. We recorded the percentage of Pareto-optimal solutions discovered by the EA for each data instance at different points of time in the execution of the EA. Then, we computed the average of the percentages across the ten data instances.

Figure 2 summarizes the results from our numerical experiment. The percentage of discovered Pareto-optimal solutions grows almost exponentially for the first 500 iterations to above 50 %, and then slows down for the remaining 1,500 iterations. At iteration 2000, when the EA stops, on the average 75 % of the Pareto-optimal solutions have been discovered by the EA. This is a very good result given that the EA finds 75 % of the Pareto-optimal solutions by searching at most 4 % of the decision space. We also notice that the performance of the EA depends on the size of the optimal-Pareto set associated with each data instance. In fact, the correlation between the size of the Pareto-optimal set and the percentage of discovered Pareto-optimal solutions by the EA is −0.77 by iteration 2000; indicating that the smaller the optimal-Pareto set, the higher the percentage of discovered Pareto-optimal solutions discovered by the EA by iteration 2000. For example, the data instance with the smallest Pareto-optimal set has 16 vectors in the set and the EA discovered 15 of those vectors by iteration 1500 (94 %).

7 Conclusion

The portfolio evaluation framework, optimization model, and corresponding solution procedure proposed in this paper add several innovations in IT PPM, including:

1. Data collection guidelines on project attributes and program status;
2. Portfolio evaluation criteria and conversion procedure from project and program attributes to model parameters;
3. Optimization engine for portfolio evaluation, project selection, and program assignment; and
4. Scenario manager supporting what-if analysis for changing project attributes, decision criteria, resource availability, and planning time horizon.

Combined, they represent an explicit road map toward three major PPM capability improvements. First, the unified evaluation framework makes it possible to assess a wide variety of projects simultaneously. Second, the evaluation can now be done in a true portfolio sense since the

impact of each project is judged not only by its own financial, strategic, and architectural contributions but also by how its inclusion or exclusion would affect others. Third, the efficient solution procedure for the portfolio optimization model generates high quality recommendations and allows for consistent and fact-based comparison among them; the decision makers can also quickly check for feasibility and optimality if there are subjective preferences or external influences that need to be accounted for. Overall, the present work is one of the very early PPM studies that use the concept of Pareto optimality to find the best mix of projects that can be completed for the given levels of resource, capability, and organizational impact.

While no automation routine could replace the experience and expertise of senior executives who make portfolio decisions, our comprehensive framework and solution procedure can be easily tailored to accommodate additional decision parameters. For example, the resource constraint in (9) can be easily changed to track multiple resource types; it is also straightforward to add dimensions to the objective function or force the selection of a project mandate. Having a decision model that closely mimics the organizational reality not only raises the decision makers' confidence in solution quality, but also helps during the conversation with functional areas as to why their proposals were not chosen.

A project could go through several rounds of portfolio reviews before its completion, so the set of projects under consideration is often a mix of new proposals (candidate projects) and on-going (active) ones. The handling of active projects is identical to that of a new project. Essentially, the project's risks, contributions, and inter-project relationship should be updated and the incremental cost (beyond what has been consumed during previous portfolio cycles) is projected using earned value analysis. Since incremental cost, not the full development cost, is used in cost profile, projects that are close to completion tend to be favored during selection. However, if there are significant changes of direction in strategy or computing infrastructure needs, then project termination is a possibility—even if the project is on time and at cost. This exemplifies the different focus between managing at portfolio versus project leveling: doing the right project precedes doing the project right.

There are several potentially fruitful extensions. The first is to investigate ways to improve the control and coordination at the program level. We have so far assumed a high level of autonomy for each program, and the decision at the portfolio level mainly is to ensure the amount of work assigned to each program matches the level of expertise and available resources. In an effort to balance workload, however, related projects may be assigned to different programs and the need for coordination may arise due to dependencies. Unlike the construction of physical artifacts that often requires stricter sequencing of tasks or projects, interrelated software projects can take place rather concurrently, provided that proper system interface definition is in place. Scheduling such projects out of sequence, however, still incurs higher overhead. For example, while it is possible to design and develop a BI reporting system's major functionality before the underlying data warehouse is in place, doing so incurs additional costs from the mock-up of technical environment and test data. Conducting testing for system performance and user acceptance may not be feasible either until after the data warehouse is online.

If it is desirable for the EPMO to also suggest the development sequence when assigning projects to programs, then existing literature on project scheduling (for extensive survey, see [20]) for such problem class (RCPSP, or, resource-constrained project scheduling problems) should be consulted. The current state of the art on RCPSP has shown to handle additional factors such as resource divisibility and sharing [36], task duration variability [12, 22], time-dependent returns [8], resource rotation [4, 37], and learning effects [17] that are common for IT projects.

Another extension will be to use the "health" of existing applications to shape project decisions. Due to changes in business processes, over time software applications invariably start losing on key functionalities or lag behind in non-functional attributes such as response time, scalability, or security. Because the integrity of the system deteriorates with each round of enhancement, a project should be planned so that a replacement can be in place before the "aging" of a system leads to its obsolescence. By observing the various indicators of an application such as the size and frequency of change requests, it is possible to determine the most cost effective point when a project should start. A properly renewed application portfolio could mean significant savings. While clearly out of scope for the current study, a co-evolution of application portfolio and project portfolio may hold the key to the IT organization's breakthrough in its value, productivity, and relevance.

References

1. Aaker DA, Tyebjee TT (1978) A model for the selection of interdependent R&D projects. IEEE Trans Eng Manage 25(2):30–36
2. Aloini D, Dulmin R, Mininno V (2007) Risk management in ERP project introduction: review of the literature. Inf Manage 44(6):547–567
3. Apfel AL (2009) Activity cycle overview: program and portfolio management. Technical report, Gartner, Inc., Report ID Number: G00167564

4. Arya A, Mittendorf B (2006) Project assignments when budget padding taints resource allocation. Manage Sci 52(9):1345–1358
5. Bardhan IR, Kauffman RJ, Naranpanawe S (2006) Optimizing an IT project portfolio with time-wise interdependencies. In: Proceedings of the 39th Hawaii International Conference on system sciences, Kauai, Hawaii, HICSS'06, p. 168b
6. Benaroch M, Kauffman RJ (1999) A case for using real options pricing analysis to evaluate information technology project investments. Inf Sys Res 10(1):70–86
7. Bohanec M, Rajkovič V, Semolić B, Pogačnik A (1995) Knowledge-based portfolio analysis for project evaluation. Inf Manag 28(5):293–302
8. Chen J, Askin RG (2009) Project selection, scheduling and resource allocation with time dependent returns. Eur J Oper Res 193(1):23–34
9. Cohen I, Golany B, Shtub A (2007) Resource allocation in stochastic, finite-capacity, multi-project systems through the cross entropy methodology. J Sched 10(3):181–193
10. Cuellar MJ, Gallivan MJ (2006) A framework for ex ante project risk assessment based on absorptive capacity. Eur J Oper Res 173(3):1123–1138
11. Deb K (2009) Multi-objective optimization using evolutionary algorithms. Wiley, New York
12. Deblaere F, Demeulemeester E, Herroelen W, Vande Vonder S (2007) Robust resource allocation decisions in resource-constrained projects. Decis Sci 38(1):5–37
13. Doerner KF, Gutjahr WJ, Hartl RF, Strauss C, Stummer C (2006) Pareto ant colony optimization with ILP preprocessing in multi-objective project portfolio selection. Eur J Oper Res 171(3): 830–841
14. Eilat H, Golany B, Shtub A (2006) Constructing and evaluating balanced portfolios of R&D projects with interactions: a DEA based methodology. Eur J Oper Res 172(3):1018–1039
15. Engwall M, Jerbrant A (2003) The resource allocation syndrome: the prime challenge of multi-project management?. Int J Proj Manage 21(6):403–409
16. Ghasemzadeh F, Archer N, Iyogun P (1999) A zero-one model for project portfolio selection and scheduling. J Oper Res Soc 50(7):745–755
17. Granot D, Zuckerman D (1991) Optimal sequencing and resource allocation in research and development projects. Manage Sci 37(2):140–156
18. Gustafsson J, Salo A (2005) Contingent portfolio programming for the management of risky projects. Oper Res 53(6):946–956
19. Heidenberger K, Stummer C (1999) Research and development project selection and resource allocation: a review of quantitative modeling approaches. Int J Manage Rev 1(2):197–224
20. Herroelen W, De Reyck B, Demeulemeester E (1998) Resource-constrained scheduling: a survey of recent developments. Comput Oper Res 25(4):279–302
21. Hess SW (1962) A dynamic programming approach to R and D budgeting and project selection. IRE Trans Eng Manag 9(4): 170–179
22. Leus R, Herroelen W (2004) Stability and resource allocation in project planning. IIE Trans 36(7):667–682
23. Levine HA (2005) Project portfolio management: a practical guide to selecting projects, managing portfolios, and maximizing benefits. Jossey-Bass, San Francisco
24. Liesiö J, Mild P, Salo A (2008) Robust portfolio modeling with incomplete cost information and project interdependencies. Eur J Oper Res 190(3):679–695
25. Linton JD, Walsh ST, Morabito J (2002) Analysis, ranking and selection of R&D projects in a portfolio. R&D Manage 32(2):139–148
26. Maizlish B, Handler R (2005) IT portfolio management step-by-step: unlocking the business value of technology. Wiley, New York
27. Man Z, Wei T, Xiang L, Lishan K (2008) Research on multi-project scheduling problem based on hybrid genetic algorithm. In 2008 International Conference on computer science and software engineering, pp 390–394
28. Mieritz L, Fitzgerald D (2010) ITS core overview for program and portfolio management. Technical report, Gartner, Inc., Report ID Number: G00205800
29. Morcos MS (2008) Modelling resource allocation of R&D project portfolios using a multi-criteria decision-making methodology. Int J Qual Reliab Manag 25(1):72–86
30. Nozick LK, Turnquist MA, Xu N (2004) Managing portfolios of projects under uncertainty. Ann Oper Res 132(1-4):243–256
31. Sanchez H, Robert B, Pellerin R (2008) Risk-opportunity identification framework. Proj Manag J 39(3):97–109
32. Sanwal A (2007) Optimizing corporate portfolio management: aligning investment proposals with organizational strategy. Wiley, New York
33. Shtub A, Bard J, Globerson S (2005) Project management: processes, methodologies, and economics. 2nd edn. Prentice-Hall, Inc., Englewood Cliffs, NJ
34. The Enterprise Portfolio Management Council (2009) Project portfolio management: a view from the management trenches. Wiley, New York
35. Wallace L, Keil M, Rai A (2004) Understanding software project risk: a cluster analysis. Inf Manag 42(1):115–125
36. Wang Y, Perkins JR, Khurana A (2002) Optimal resource allocation in new product development projects: a control-theoretic approach. IEEE Trans Autom Control 47(8):1267–1276
37. Yang K-K, Sum C-C (1997) An evaluation of due date, resource allocation, project release, and activity scheduling rules in a multiproject environment. Eur J Oper Res 103(1):139–154